

KAILASH AMBARAM CHAUDHARY

Python/Backend Developer | Cloud & DevOps Engineer | AI/ML Engineer

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PROFESSIONAL SUMMARY

2025 Computer Science Engineering graduate with hands-on internship experience across AWS cloud infrastructure, DevOps automation, and applied AI/ML. Built and deployed CI/CD pipelines (GitHub Actions, Jenkins, CodeDeploy), provisioned infrastructure as code with Terraform, and containerized applications using Docker and Kubernetes. Developed a production-style Retrieval-Augmented Generation (RAG) system using FastAPI, FAISS, and LLM inference, and a deep-learning video activity recognition pipeline (CNN+LSTM) with a published research paper. Strong builder mindset with a track record of independently shipping full-stack, cloud-deployed projects from scratch. Comfortable across Python backend development, cloud operations, and ML pipelines, with AWS and IIT Roorkee DevOps certifications.

TECHNICAL SKILLS

- **Programming & Backend:** Python, FastAPI, Flask, REST APIs, SQLAlchemy, MySQL, Pydantic, Gunicorn
- **Cloud & DevOps:** AWS (EC2, S3, IAM, VPC, ALB, ASG, CodeDeploy, CloudWatch, WAF, Boto3, CLI), Terraform, Docker, Kubernetes, GitHub Actions, Jenkins, GitLab CI (familiar), Linux administration
- **AI/ML & Data:** TensorFlow, Keras, OpenCV, LangChain, FAISS, HuggingFace Transformers, Groq/Llama 3.1, RAG pipelines, LSTM, CNN, NumPy, Pandas, Scikit-Learn, Streamlit
- **Monitoring & Reliability:** CloudWatch dashboards & alarms, SNS notifications, log-based RCA, runbook authoring, ITSM concepts (Jira, ServiceNow)
- **Tools:** Git/GitHub, Docker Compose, YAML, Bash/Shell scripting, Jupyter, VS Code

WORK EXPERIENCE

Cloud & DevOps Intern — TechEazy Consulting

Oct 2025 – Dec 2025

- Monitored AWS EC2 fleets using CloudWatch dashboards and alarms, tracking CPU utilisation, network traffic, and error rates in real time; configured SNS notifications for threshold breaches to support fast incident response.
- Diagnosed and resolved IAM permission boundary errors and CI/CD pipeline (YAML) misconfigurations through systematic log analysis; authored RCA documentation and deployment runbooks adopted by the team.
- Provisioned and maintained AWS infrastructure (EC2, S3, IAM, ALB, Auto Scaling Groups) using Terraform and Boto3, replacing manual console steps with version-controlled, repeatable infrastructure.
- Configured AWS WAF rate-limiting and geo-blocking rules to reduce malicious traffic and improve monitoring signal-to-noise ratio.
- Collaborated with development and QA teams in an Agile workflow, contributing to sprint goals around deployment reliability and operational documentation quality.

Cloud Associate Engineer Intern — TechnoGrowth Software Solutions

Dec 2023 – Jan 2024

- Configured and maintained Docker containers and Linux-based server environments, supporting consistent, reproducible deployments across development and staging.
- Performed post-release deployment validation checks and authored onboarding documentation and knowledge-base articles for new team members.

PROJECTS

LLM Document Q&A System (RAG Pipeline) | Python · FastAPI · LangChain · FAISS · Groq Llama 3.1 · Docker · Kubernetes | github.com/kailashji24/llm-document-q-a-rag

- Built a Retrieval-Augmented Generation pipeline for PDF document Q&A: ingestion, chunking, HuggingFace embeddings, FAISS vector search, and context-grounded answer generation via Groq-hosted Llama 3.1.
- Exposed the pipeline as a FastAPI microservice with health-check endpoints; deployed to Kubernetes using Deployments, Services, ConfigMaps, and Secrets with readiness/liveness probes.
- Resolved FAISS index-availability and pod networking issues inside Kubernetes; automated build/validation through a GitHub Actions CI pipeline.

Automated CI/CD Pipeline — AWS Lift & Shift | Terraform · GitHub Actions · AWS EC2/ALB/ASG/S3/CodeDeploy/CloudWatch | github.com/kailashji24/test-repo-for-devops-app

- Provisioned complete AWS infrastructure as code using Terraform (VPC, subnets, security groups, ALB, ASG, IAM roles) — zero manual console steps, fully repeatable.
- Automated the build-to-deploy lifecycle via GitHub Actions and CodeDeploy; diagnosed and fixed IAM boundary conflicts and lifecycle-hook failures through CloudWatch log analysis.
- Configured CloudWatch dashboards and SNS alarms to monitor deployment health, instance metrics, and ALB request counts for proactive incident response.

Suspicious Activity Detection System | *Python · TensorFlow · Keras · OpenCV · LSTM/CNN · Streamlit* | github.com/kailashji24/Suspicious-Activity-Detection

- Built an LRCN (CNN+LSTM) deep-learning pipeline to classify human activities (Walking, Running, Fighting) from surveillance video frames, combining CNN spatial feature extraction with LSTM temporal sequence modeling.
- Deployed the trained model as an interactive Streamlit dashboard with video upload, confidence scoring, and prediction logging/analytics.
- Published associated research: “Investigation on Human Activity Recognition using Deep Learning,” IRJMETS, Volume 07, Issue 05, May 2025.

Student Registration DevOps Platform | *Python · Flask · SQLAlchemy · MySQL · Docker Compose · Jenkins* | github.com/kailashji24/flask-student-registration-devops

- Developed a Flask + SQLAlchemy + MySQL CRUD application for student registration management; containerized with Docker and orchestrated as a multi-service stack using Docker Compose with persistent volumes and health checks.
- Built a Jenkins CI/CD pipeline to automate build, validation, and redeployment on each code push; deployed the app using Gunicorn as the production WSGI server.

EDUCATION

B.E., Computer Science Engineering — CGPA: 7.90 / 10
PREC, Savitribai Phule Pune University, Loni, Maharashtra

2025

CERTIFICATIONS

- Advanced Certification in Cloud Computing and DevOps — iHUB DivyaSampark, IIT Roorkee, in association with Dept. of Science & Technology, Govt. of India (Jul 2025) — ID: IPTIH25070274
- AWS Solutions Architect Associate — Certificate of Completion, Intellipaat (Mar 2025) — ID: 31679-13867-284472
- AWS Solutions Architecture Job Simulation — Forage (Jul 2025)

PUBLICATION

- “Investigation on Human Activity Recognition using Deep Learning,” International Research Journal of Modernization in Engineering Technology and Science (IRJMETS), Volume 07, Issue 05, May 2025 (e-ISSN: 2582-5208, Impact Factor 8.187).

ADDITIONAL

- Open to 24/7 rotational shifts and on-call support rotations. Actively practising Data Structures & Algorithms on LeetCode.
- All projects hosted on GitHub with structured documentation and runbook-style READMEs.